NAU NORTHERN ARIZONA UNIVERSITY
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Enumeration of Signed Permutations Under the Action of Reversals

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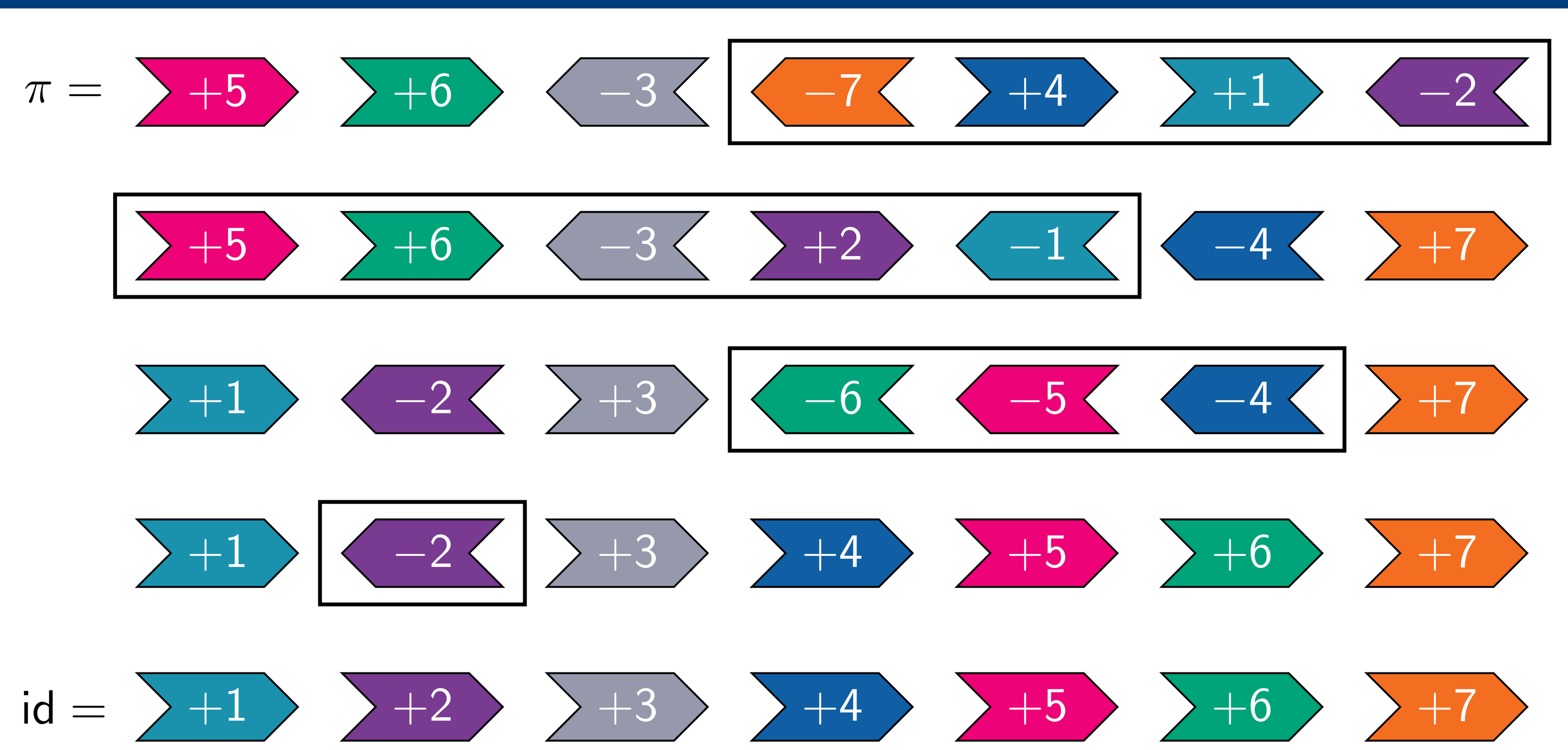
Biology to Mathematics

- The **edit distance** between two genomes is the minimum number of mutations required to transform one into another. This approximates evolutionary distance.
  - mouse $\xrightarrow{251}$ human (149 reversals)
  - cabbage $\xrightarrow{3}$ turnip (all reversals)
- Comparing two similar sequences of genes appearing along a chromosome in two species yields two **signed permutations**.
- **Reversal distance** between two signed permutations is the minimum number of reversals necessary to transform one into the other. Good estimate of evolutionary distance.

Signed Permutations and Reversals

- **Permutation:** A list of numbers 1 through n with a specific ordering. For example, [3, 2, 4, 5, 1] is a permutation of 1 through 5.
- **Signed Permutation:** A permutation of numbers 1 through n where each number is signed. For example, [4, -5, -2, -1, -3] is a signed permutation of 1 through 5. The collection of all signed permutations is denoted $S_n^{\pm}$.
- **Reversal:** The act of swapping the order of a consecutive subsequence and changing the sign of each number in the subsequence. We denote a reversal on $\pi = [\pi_1, \pi_2, \dots, \pi_n]$ by $\delta_{i,j}$, where $\delta_{i,j}$ reverses and changes the signs of elements $\pi_i, \pi_{i+1}, \dots, \pi_{j-1}, \pi_j$.
- **Breakpoint Diagram:** Is a graph constructed from a signed permutation π denoted by $BG(\pi)$. It provides useful information to calculate the reversal distance of π.
- Define $d_{\text{rev}}(\pi)$ of a signed permutation π to be the minimum number of reversals needed to transform π into the identity permutation.
- Define $R_{n,k} := \{\pi \in S_n^{\pm} \mid d_{rev}(\pi) = k\}$ to be the set of signed permutations with reversal distance k and define $r_{n,k} := |R_{n,k}|$.

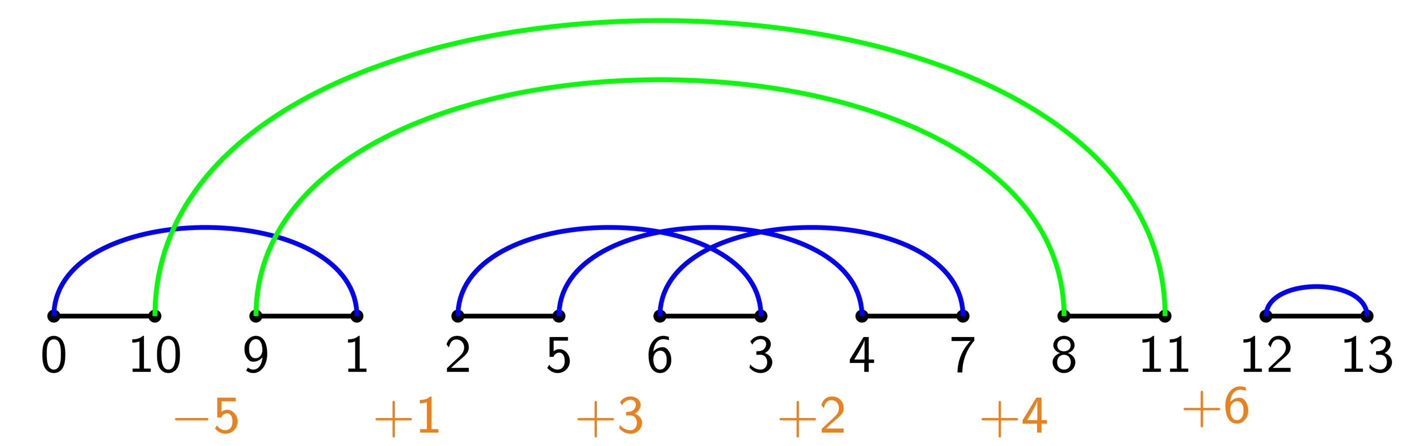
Example



$d_{\text{rev}}(\pi) \le 4$

Breakpoint Diagrams

Every permutation has a unique **breakpoint diagram**



Features of Breakpoint Diagrams

- **Cycle:** A closed loop connected via arcs and black edges on the breakpoint diagram. A cycle is called **oriented** if an arc connects a positive integer to a negative integer and **unoriented** otherwise. The cycle with the green arc is oriented while the cycle with only blue arcs is unoriented.
- **Trivial Cycle:** A cycle that is composed of a single arc and a single black edge. E.g. The last cycle in the example breakpoint diagram above.
- **Hurdle:** A collection of overlapping unoriented cycles that either covers all other collections of unoriented cycles or covers no collections of unoriented cycles.
- **Superhurdle:** A hurdle that if removed creates a new hurdle.
- **Fortress:** A permutation with an odd number of hurdles where every hurdle is a superhurdle. We have show that fortresses only occur in $\pi \in S_n^{\pm}$ where n is odd and $n \ge 17$.

Reversal Distance of a Signed Permutation

Theorem (Hannenhalli & Pevsner 1999)
The reversal distance for a signed permutation π of length n is

$$d_{\text{rev}}(\pi) = n + 1 - c(\pi) + h(\pi) + f(\pi),$$

where $c(\pi)$ is the number of cycles in $BG(\pi)$, $h(\pi)$ is the number of hurdles in $BG(\pi)$, and $f(\pi) = 1$ if π is a fortress and 0 otherwise.

Maximal Reversal Distance

Theorem
The maximal reversal distance for signed permutations $\pi \in S_n^{\pm}$ is given by

$$d_{rev}^{\max}(S_n^{\pm}) = \begin{cases} n, & \text{if } n \in \{1,3\} \\ n+1, & \text{otherwise} \end{cases}$$

The following formula enumerates the number of permutations with maximal reversal distance.

Number of Maximal Permutations

Theorem (Awik, Ernst, Rosenberg 2021)
The number of maximal permutations $r_{n,n+1}$ for $n \ge 4$ is:

$$r_{n,n+1} = \sum_{(\alpha_1,\dots,\alpha_k)\in C_{odd}^{>1}} \left(\prod_{i=1}^{k} \frac{2(\alpha_i - 1)!}{\alpha_i + 1} \right) \cdot \begin{cases} \alpha_1, & \text{if } k \ne 1 \\ 1, & \text{if } k = 1. \end{cases}$$

where $C_{odd}^{>1}$ denotes the set of compositions of $n+1$ where each α_i is odd and greater than 1.

Distribution of Maximal Signed permutations

Conjecture (Denoncourt, Ernst, Rosenberg 2020)

$$\lim_{n\to\infty} \frac{r_{n,n+1}}{2(n-1)!} = 1 \quad \text{if } n \text{ is odd}$$
$$\lim_{n\to\infty} \frac{r_{n,n+1}}{2(n-3)!} = 1 \quad \text{if } n \text{ is even}$$

This implies that choosing a maximal permutation from $S_n^{\pm}$ becomes more unlikely as n grows.

General Enumerative Formula

Theorem (Burkhart, Ernst, Stewart 2022)
The number of signed permutations in $S^{\pm}(n)$ with reversal distance k is given by

$$r_{n,k} = a_{k-1,k}\binom{n+1}{k} + a_{k,k}\binom{n+1}{k+1} + \cdots + a_{2k-1,k}\binom{n+1}{2k}$$

where each coefficient $a_{i,k}$ for $k-1 \le i \le 2k-1$ is the number of signed permutations without trivial cycles in their breakpoint diagrams in $S_i^{\pm}$ and the binomial counts the number of ways to place the signed permutation without trivial cycles onto $i+1$ of the $n+1$ black edges.

- It turns out that $r_{n,k}$ is a polynomial of degree $2k$.

Number of Lower Distance Permutations

Theorem (Awik, Ernst, Rosenberg 2021)
The number of signed permutations with reversal distance 1 and 2 are given by

$$r_{n,1} = \frac{n(n+1)}{2} \quad \text{and} \quad r_{n,2} = \frac{(n-1)n(n+1)^2}{6}.$$

Theorem (Burkhart, Ernst, Stewart 2022)
The number permutations with reversal distance 3 and 4 are given by

$$r_{n,3} = \frac{(n-1)n^2(n+2)(7n-11)}{144}$$

$$r_{n,4} = \frac{(n-3)(n-2)(n-1)n(n+1)(37n^3 + 103n^2 + 8n + 12)}{3360}$$

$$r_{n,5} = 8\binom{n}{5} + 1399\binom{n}{6} + 9534\binom{n}{7} + 22337\binom{n}{8} + 21744\binom{n}{9} + 7534\binom{n}{10}$$

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